Assignment 1

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Question 1

1.1 Detect how much stETH depeg to $ETH on Ethereum main network.

I have obtained the price data and liquidity data for this pool during the period from August 4 to 6, 2024. The data is stored in a CSV file named steth_weth_uniswapv2_20240804_06. The chart generated from the data is shown in Figure 1.

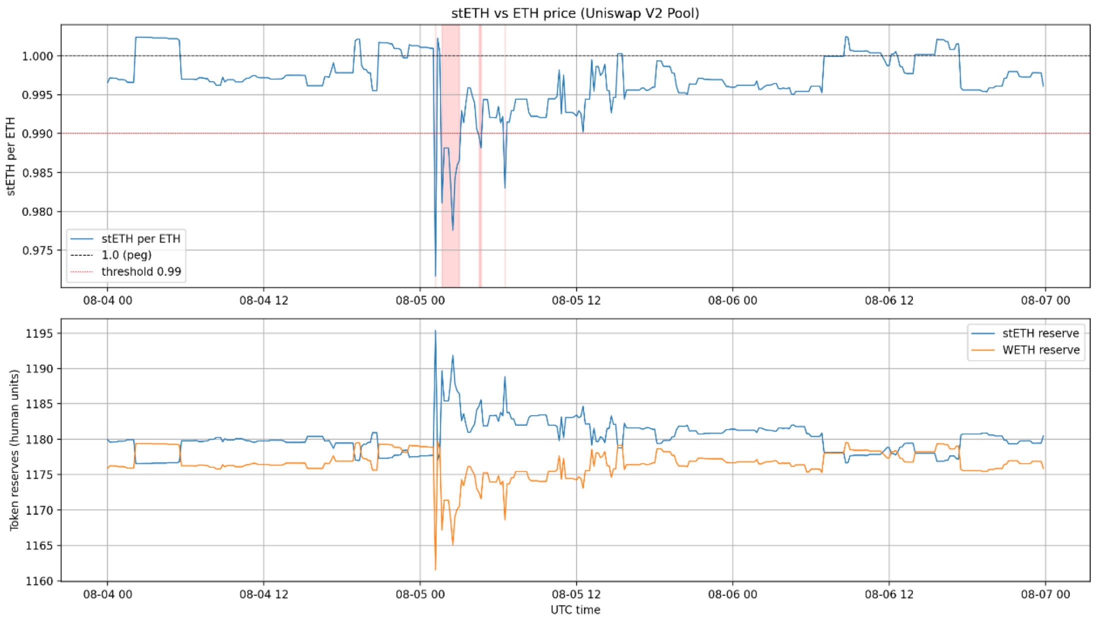

Figure 1 ETH-ezETH Liquidity Pool Data Changes

As shown in the Figure 2, during the early hours of August 5 (red shaded area), massive market sell-offs of stETH caused a surge in the pool's stETH supply. This triggered a sharp decline in stETH's price, which briefly fell below 0.98, deviating significantly from its peg. The price subsequently recovered gradually and later returned to a level close to 1. This is consistent with the events that occurred in history.

1.2 Detect how much ezETH depeg to $ETH on Ethereum main network.

Similar to the previous question, I obtained the relevant transaction data and saved it in a CSV file named ezeth_weth_uniswapv3_20240804_06.

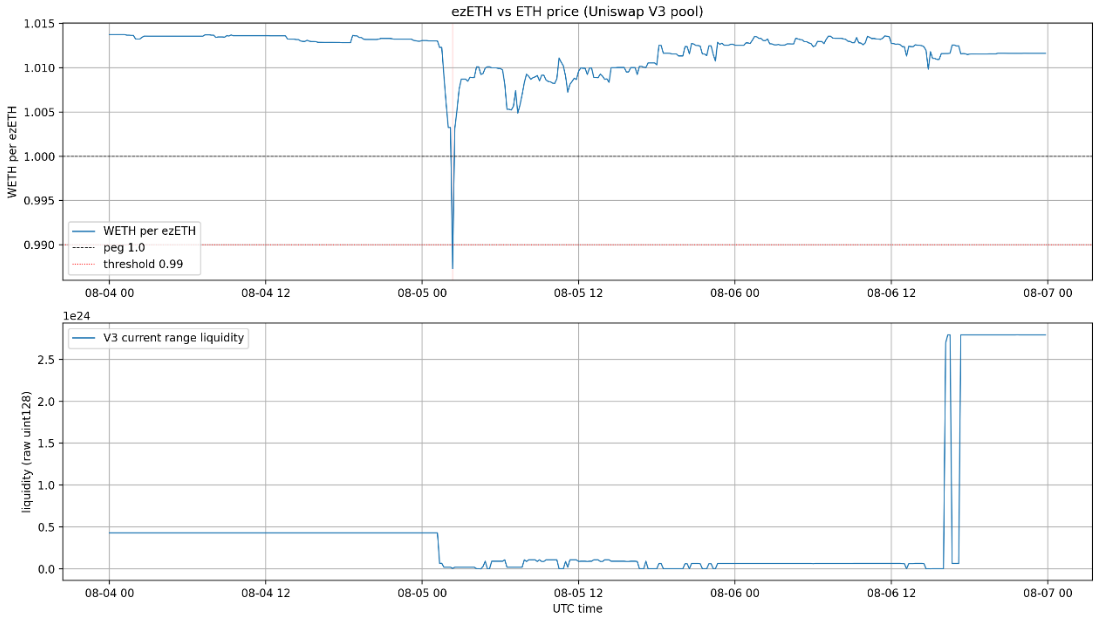

Figure 2 ETH-ezETH Liquidity Pool Data Changes

This chart illustrates the price and liquidity changes of ezETH relative to ETH within the Uniswap V3 pool: Prior to the early hours of August 5th, ezETH's price remained slightly above 1 (pegged to 1) against ETH with stable liquidity. During the early hours of August 5th, ezETH's price plummeted sharply (briefly deviating from the peg), accompanied by a significant decline in pool liquidity. Subsequently, ezETH's price gradually recovered, and on the evening of August 6th, the pool experienced a substantial influx of liquidity.

Question 2

2.1 What token and how much of that token is being taken out of the pool ? and What token and how much of that token is being sent into the pool ?

In this transaction, targeting the WETH/USDC pool on Uniswap V2 (pool contract address: 0xB4e16d0168e52d35CaCD2c6185b44281Ec28C9Dc), logs indicate the trader (via Uniswap Router) **sent approximately 0.59093884 WETH into the pool** (as input) and **withdrew approximately 1,024.229 USDC from the pool (as output)**. In other words, the trader exchanged approximately 0.59094 WETH for approximately 1,024.23 USDC.

2.2 What is the position of this transaction within that block ? How does transaction gets their order no within the block and why does it matter ?

This transaction is located at block 17462685 with transactionIndex = 27 (i.e., the 28th transaction in this block). The final order of transactions within a block is determined by the block producer during block construction (based on factors such as the mempool, transaction fees, nonce constraints, gas limits, and MEV trade-offs). Once the block is finalized, this order is recorded on-chain and becomes definitive. Transaction order is critically important because sequential placement alters contract state (e.g., AMM pool reserves), thereby influencing subsequent transaction outcomes, actual execution prices, and slippage. This creates risks of front/sandwich attacks (MEV) and transaction failures.

2.3 Write a python code to simulate this transaction

The code is stored in a file named uniswap_v2_sim.py. I'll demonstrate the output assuming pool reserve0=500 stETH, reserve1=1000 WETH, and a user swaps 0.5 token0 (token0 -> token1) with a 0.3% fee. The output is shown in Figure 3.

```
D:\pythoneviroment\python.exe D:\NTU\Semester1\S6124\Assignment1\uniswap_v2_sim.py
Simulation example:
  token_in: token0
  token_out: token1
  amount_in_raw: 0.5
  fee: 0.003
  amount_in_with_fee: 0.4985
  reserve_in: 500.0
  reserve_out: 1000.0
  numerator: 498.5
  denominator: 500.4985
  amount_out: 0.9960069810399033
  => amount_out = 0.996006981040 token1
```

Figure 3 Question2.3 Result

2.4 Test your code by putting input of the transactions in the function. Note: it is expected that the input&output will match exactly the trans action above

To address question 2.4, I made the following modifications to the code in question 2.3:
1. Retrieve the corresponding Pair address via RPC.
2. Read the getReserves() call for that Pair in the block preceding the transaction (block_number - 1) to obtain the precise pre-transaction reserves .
3. Determine token0/token1 and their respective decimals, then convert amount_in to an integer using the corresponding decimals.
4. Precisely calculate amount_out_raw using the Uniswap V2 formula to ensure consistency with the on-chain implementation.

The code for this problem is stored in the file uniswap_v2_q2_4.py. The output of the code is shown below.

```
D:\pythoneviroment\python.exe D:\NTU\Semester1\S6124\Assignment1\uniswap_v2_q2_4.py
Transaction block: 17462685 tx index: 27
Pair token0: 0xA0b86991c6218b36c1d19D4a2e9Eb0cE3606eB48
Pair token1: 0xC02aaA39b223FE8D0A0e5C4F27eAD9083C756Cc2
Reserves at pre-swap block 17462684: reserve0=26853692230452, reserve1=15446428142167535900067
WETH is token1 in this pair.
reserve_in: 15446428142167535900067
reserve_out: 26853692230452

Simulation (integer math):
 amount_in_raw: 590938840873296854
 amount_in_with_fee (raw*997): 589166024350676963438
 numerator: 15821283090552067811321520674213976
 denominator: 15447017308191886577030438
 simulated amount_out (raw): 1024229000
 on-chain amount_out (raw from earlier decode): 1024229000

Human readable:
 Input: 0.590938840873296845 WETH
 On-chain output: 1024.229000 USDC
 Simulated output: 1024.229000 USDC

RESULT: Simulation MATCHES on-chain amount_out exactly.
```

Figure 4 Question2.3 Result